

## GIF Maker

### 1. Description

This implementation for Unity3D allows you to record and save GIF animations in runtime. It will be very useful for creating small GIF animations of gameplay that may be shared on the web.

It uses RenderTexture component to capture the frames and NGif for encoding GIF files. NGif based on LZW image compression and Neural Image Quantization algorithms which runs in thread for better performance. Working on standalone and mobile platforms.

**NOTE! It uses FileStream class so you won't be able to save GIF files from Web Player.**

Features:

- \* Easy! All you need is three methods from GifMaker class: StartRecord, StopRecord and SaveGif
- \* Flexible configuration(delay, repeat count, quality, size)
- \* Encoding process runs in thread for better performance

Look at commented *GifMakerExample.cs* script to see how GIF Maker should be configured and running.

## 2. How to use

A. First of all you need to call **GifMaker.StartRecord()** method for start capturing frames. It takes 7 arguments that as described below.

You can assign more than one camera for capturing also UI layer for example. In additional, `OnStartFrameCapture` and `OnEndFrameCapture` can be called if needed to change something on the screen before capturing the frame.

```
public void StartRecord()
{
    GifMaker.GifRecordOptions recordOptions = new GifMaker.GifRecordOptions(
        Camera.main, //assign main camera for capturing frames
        0.1f, //capture 10 frames every second
        new Rect(0, 0, Screen.width, Screen.height), //record all area of the camera
        256, //resize frame to 256 by width
        256, //resize frame to 256 by height
        OnStartFrameCapture, //assign OnStartFrameCapture method
        OnEndFrameCapture); //assign OnEndFrameCapture method

    GifMaker.StartRecord(recordOptions);
}
```

B. Then, call **GifMaker.StopRecord()** method for stop capturing frames. Do not record too many frames because they stored in memory in Texture2D format.

```
public void StopRecord()
{
    GifMaker.StopRecord();
}
```

C. Finally, call **GifMaker.SaveGif()** to run encoding thread. Duration of the encoding process based on size of the frames and how many frames has been recorded.

```
public void SaveGif()
{
    GifMaker.GifEncodeOptions encodeOptions = new GifMaker.GifEncodeOptions(
        Application.persistentDataPath + "\\Test.gif", //Set destination folder for gif
        animation file
        0, //Set animation repeat count to loop
        1, //Set the best possible quality for gif
        animation
        0.1f, //Set delay for animation to 10 frames
        per second
        OnSaveComplete); //Assign OnSaveComplete method

    GifMaker.SaveGif(encodeOptions);
}
```